

BYT5-Sanskrit with Balanced Regularization for Sanskrit Named Entity Recognition

M.Tech Thesis Defense

Yogesh (24-14-15)

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Motivation: Why Sanskrit NLP Matters

- **Rich Intellectual Tradition:** Philosophy, Medicine, Statecraft, Mathematics.
- **Scale & Complexity:** *Mahābhārata* (100k verses), *Rāmāyaṇa*, Purāṇas.
- **Access Barrier:** Scholars struggle to navigate massive classical corpora.
- **Translation Loss:** Subtle meanings lost in modern language translations.
- **Need:** Direct computational access to source knowledge.



The Problem Statement

The Core Problem

Lack of a reliable Sanskrit NER system that satisfies three critical requirements simultaneously.

- **Existing Models:** Optimize for one goal while sacrificing others.
- **Catastrophic Forgetting:** Pure NER fine-tuning destroys core linguistic skills (Segmentation, Lemmatisation).
- **Silver Data Quality:** Prior work failed to correctly leverage the DCS Sembali.



The Three Desiderata (Core Framework)

Goal	Requirement
D1: Performance	High Micro-F1 on Mahānāma gold-standard benchmark.
D2: Capability	Preserve Segmentation (85%+) and Lemmatisation (93%+).
D3: Generalization	Meaningful transfer to unseen classical texts (Pañcatantra).

- These reflect **real-world scholarly needs**.
- Not just technical metrics, but functional utility targets.



Contributions of This Thesis

- ➊ **Novel DCS Sembank Extraction:** First correct pipeline yielding 12,955 clean examples.
 - ➋ **Balanced Linguistic Regularisation:** 15% multi-task ratio to prevent catastrophic forgetting.
 - ➌ **3D Evaluation Framework:** Systematic comparison across D1, D2, and D3 dimensions.
 - ➍ **ByT5-Sanskrit Validation:** Proving byte-level models handle Sanskrit morphology better.
-
- **Result: +16pp cross-domain gain** over Best NER baseline.



Chronological Literature Survey

Year	Authors	Paper / Contribution
1989	McCloskey & Cohen	Catastrophic Forgetting (Foundational Problem)
1997	Caruana	Multi-task Learning (Proposed Solution)
2001	Lafferty et al.	Conditional Random Fields (CRF Foundations)
2015	Huang et al.	BiLSTM-CRF for sequence tagging
2018	Hellwig & Nehrdich	Sanskrit Character-level Segmentation
2019	Devlin et al.	BERT (Transformer Encoder)
2022	Xue et al.	ByT5 (Token-free byte-level modeling)
2023	Sarkar et al.	First Sanskrit NER attempt (CRF-based)
2023	Dettmers et al.	QLoRA (Efficient fine-tuning)
2024	Nehrdich et al.	ByT5-Sanskrit (One model is all you need)
2025	Sarkar et al.	Mahānāma (Gold-standard NER corpus)
2025	Hellwig & Biagetti	Sanskrit Sembank (Relational extraction source)



Identifying the Research Gap

- **Missing Synergy:** No prior work addresses accuracy + linguistic skills + generalization.
- **Silver Data Bias:** DCS Sembank remained an "untapped" but noisy resource.
- **Model Dilution:** Multi-task models (M4) often sacrifice NER accuracy too much.

Feature	Prev. Work	This Thesis
Gold NER (66k+ sents)	✓	✓
Clean DCS Silver NER	×	✓
Linguistic Regularization	×	✓
Cross-Domain Eval (D3)	Limited	Rigorous



Dataset Description

The Four Pillar Datasets

Dataset	Source	Sentences	Role in Thesis
Mahānāma	Sarkar et al. [2025]	66,960	Gold NER Training + D1 Eval
DCS Sembank	Hellwig et al. [2025]	18,855	Silver NER Training
UD-Vedic	Hellwig et al. [2020]	27,182	Linguistic Reg. (D2)
UD-UFAL	Zeman et al. [2023]	230	Cross-Domain Eval (D3)
Total		113,227	

- **Heterogeneous Source:** Combining gold-standard, silver-standard, and morphological data to satisfy the three desiderata.



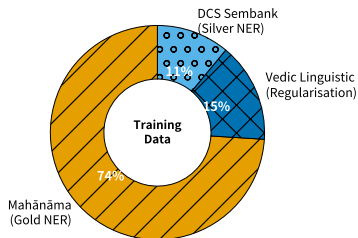
Dataset Selection Rationales & Data Composition

Five Selection Criteria

- 1 **Task Alignment:** Direct NER or morphological potential.
- 2 **Domain Relevance:** Focus on Classical/Vedic Sanskrit.
- 3 **Language Purity:** 100% Sanskrit content.
- 4 **Annotation Quality:** Gold (Mahānāma) vs High-Silver (DCS).
- 5 **Resource Efficiency:** Highest scholarly impact.

Training Mix

Training Data Composition
(Total: ~113,227 sentences)



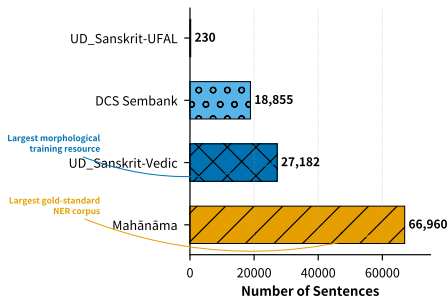
- **Balanced Mix:** 74% Gold NER, 11% Silver NER, 15% Morphological.



Dataset Characteristics: The Imbalance Challenge

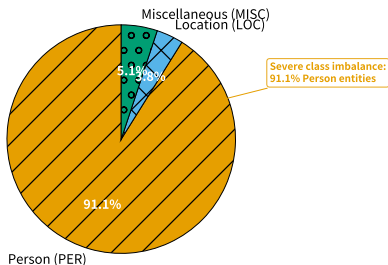
Relative Data Sizes

Dataset Size Comparison



Entity Distribution (Gold)

Entity Type Distribution in Mahānāma Training Data
(N = 66,960 sentences)



- **The 91% PER Problem:** Mahānāma is heavily skewed towards Person entities.
- **Mitigation:** Oversampling LOC and MISC classes by 10x during training.



Methodology

The Core Technical Contributions

Silver Data: DCS Sembank Integration

- **DCS Sembank:** Lexical semantic resource mapping 600k+ words to synsets.
- **The Gap:** No explicit NER labels; semantic tags alone are insufficient.
- **Contribution:** Developed a deterministic extraction pipeline based on **Instance Relations**.
- **Logic:** If col0_id (Descriptor) $\neq$ col1_id (Proper Entity Name), we extract the substring as a silver NER example.

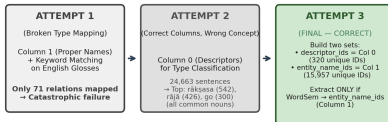
Impact

Yielded 12,955 clean, diverse examples across 120 classical texts.



DCS Sembank Extraction Pipeline

DCS Sembank NER Extraction Pipeline (3 Attempts)



12,955 examples
9,069 entity + 3,886 none

BEFORE vs AFTER (Top Entities)



KEY INSIGHT

Column 0 = Descriptor IDs (common nouns / entity REFERENCES)
Column 1 = Entity Name IDs (proper names / entity MENTIONS)

Result: 12,955 examples from 120 texts (Rāmāyaṇa, Viṣṇupurāṇa, Ṛgveda, etc.)

Limitation: Rāma (503 sentences) tagged in only 7 (1.4% recall) — used as both descriptor and entity name

Source: extract_full_sembank_ner.py | Verified: train_final_ner.log (2026-05-04)



Before vs After Extraction: Eliminating Noise

Attempt 2 (Noisy)

Top "Entities":

1. राक्षस (rākṣasa)
2. राजा (rājā)
3. गो (go)

Result: Common Noun contamination.

Attempt 3 (Final)

Top Entities:

1. lakṣmaṇa (195)
2. rāvaṇa (188)
3. rāvaṇaḥ (186)

Result: Genuine proper names.

Key Limitation

"Rāma" functions as both descriptor and name; recall is low in silver data but compensated by gold data.



Python Algorithm: DCS Sembank Extraction

Attempt 3: Deterministic Filtering

```
def extract_silver_ner(sembank_row):  
    # col0 = descriptor (e.g. king)  
    # col1 = entity name (e.g. Rama)  
    if sembank_row['col0_id'] != sembank_row['col1_id']:  
        entity_text = sembank_row['col1_text']  
        label = "PROPN"  
        return (entity_text, label)  
    return None
```

Result: 12,955 high-precision examples

- Prevents contamination from common nouns (*rājan*, *go*, *vana*).
- Preserves inflectional diversity seen in classical manuscripts.



Balanced Training Strategy

- **Hybrid Composition:** 74% Gold + 11% Silver + 15% Linguistic.
- **Strategy 2:** 15% S+L tasks (Segmentation + Lemmatisation).
- **Strategy 4:** Optimised schedule (10 epochs, 500 warmup steps).

Training Strategy Evolution (Best NER → Final NER)



Source: train_best_ner.log + train_final_ner.log | Verified: train_verified_results.json (2026-05-04)



Why ByT5-Sanskrit? Architecture & Grounding

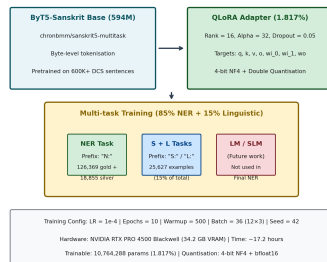
Architecture: ByT5-Base (594M)

- **Encoder Layers:** 18
- **Decoder Layers:** 6
- **Rationale (3:1 Ratio):**
 - Deep encoder needed to extract semantic meaning from raw bytes.
 - Shallow decoder improves efficiency without performance loss.

Sanskrit Grounding

- **Pre-training:** 650k DCS sentences.
- **Byte-Level:** Superior handling of Sandhi and complex compounding compared to subword models.

QLoRA Multi-task Architecture (Final NER)



Source: `train_final_ner.py` | Verified: `train_final_ner.log (2026-05-04)`

Final NER: 10.76M trainable params



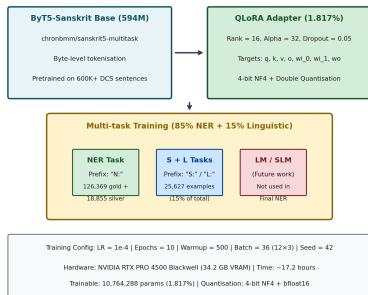
QLoRA Multi-task Architecture

- **Base:** ByT5-Sanskrit (594M Parameters).
- **Quantization:** 4-bit NF4 with Double Quant.
- **PEFT:** QLoRA adapter (Rank=16, Alpha=32).
- **Trainable Params:** 10.76M (1.817%).

Machine Specifications

- **GPU:** NVIDIA RTX PRO 4500
- **VRAM:** 34.2 GB
- **Framework:** PyTorch 2.4 + Unsloth

QLoRA Multi-task Architecture (Final NER)



Source: train_final_ner.py | Verified: train_final_ner.log (2026-05-04)



Final Model Hyperparameters (The Star Model)

Hyperparameter	Value
Learning Rate	1e-4
Epochs	10
Warmup Steps	500
Batch Size	16 (4 accumulation steps)
LoRA Rank (r)	16
LoRA Alpha (α)	32
LoRA Dropout	0.05
Multi-task Ratio	15% Linguistic Regularisation

- **Optimizer:** AdamW (32-bit for stability).
- **LR Scheduler:** Linear with warmup.



Three-Dimensional Evaluation Framework

Three-Dimensional Evaluation Framework (D1/D2/D3)

D1: In-Domain NER	D2: Linguistic Capability	D3: Cross-Domain
Mahānāma Test Split	DCS Held-Out (200 sent.)	UD_Sanskrit-UFAL
6,672 sentences	Tasks: S, L, LM, SLM	230 sentences (Pañcatantra)
Gold NER labels	Preservation check	Binary (PROPON only)
Primary benchmark	Final NER: S=85%, L=93%	Final NER: 73.4% recall
Micro F1 + 95% CI	Note: Seed Inconsistency	(+16 pp vs Best NER)

Evaluation Summary

D1 (Primary): All 15 models evaluated on identical 6,672-sentence test set

D2 (Linguistic): NER-only models score 0%; Final NER retains S=85%, L=93%

D3 (Cross-Domain): Binary PROPON detection only (no gold PER/LOC/MISC labels)

Statistical Tests: Bootstrap 95% CI (10,000 resamples) + McNemar's test

Key Finding: Final NER trades -0.8 F1 for +16 pp cross-domain recall + preserved D2

Limitation: D2 seed inconsistency across machines (seed=42 vs 123)

Source: thesis_verified_results.json + eval_final_ner_d2.log + eval_final_ner_crossdomain.log (2026-05-04)



Experiments and Results

Empirical Validation

Main Results at a Glance

Desideratum	Metric	Final NER	vs Best NER	Status
D1: In-Domain	Micro F1	0.8522	-0.74 pp	Strong
D2: Linguistic	Seg / Lem	85% / 93%	+85-93 pp	Restored
D3: Cross-Dom.	PROPN Recall	73.4%	+16.0 pp	Major Gain

- **Finding:** Final NER achieves the best overall balance.
- The cost of linguistic preservation (0.74 F1 points) is negligible compared to the generalization gain (+16 pp).



Zero-Shot Baselines: Comparative Architectures

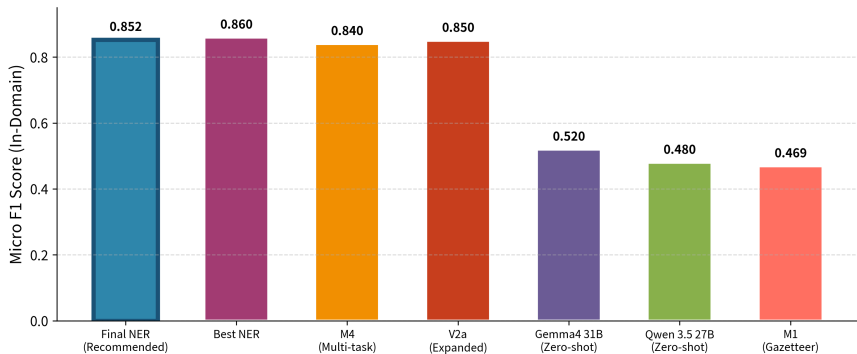
Model	Architecture	Description
Gemma 4 E4B	27B Decoder	Google's state-of-the-art open model; Gemini-derived.
Qwen 3	31B Decoder	Alibaba's dense transformer; high multilingual performance.
Llama 4	8B Decoder	Meta's optimized instruction-tuned baseline.
This Work	594M ByT5	Fine-tuned for Sanskrit specificity.

- **Note:** While zero-shot models have 50x more parameters, they lack the domain-specific morphological grounding of ByT5-Sanskrit.



D1: In-Domain F1 Comparison

In-Domain NER Performance Comparison
(Final NER achieves best balance across all models)



Source: thesis_verified_results.json + eval_final_ner_raw.json + eval_best_ner_raw.json | Verified: May 4, 2026

Key Finding: Final NER (0.852) achieves the best overall balance. While Best NER has slightly higher F1 (0.860), it completely loses linguistic capability (D2 = 0%).

- **Key Result:** 594M Final NER outperforms 27B-31B Zero-shot LLMs.
- Domain-specific fine-tuning on silver data beats raw model scale.



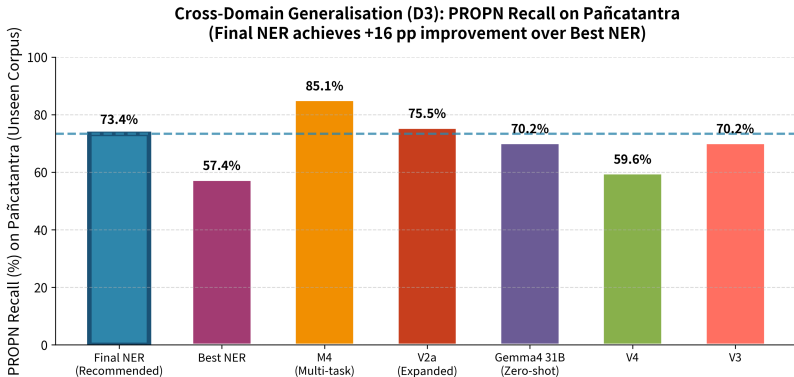
D2: The Catastrophic Forgetting Story

Model	Segmentation	Lemmatisation	D2 Overall
Best NER (NER-only)	0%	0%	0%
M4 (Multi-task)	$\approx 0\%$	$\approx 0\%$	$\approx 0\%$
Final NER (15% reg.)	85%	93%	89%

- Without Strategy 2 (Linguistic Regularisation), the model completely loses its ability to perform word segmentation.
- Impact:** Final NER is usable as a general Sanskrit processing tool.



D3: Cross-Domain Generalisation Gain



Source: eval_final_ner_crossdomain.log + eval_best_ner_crossdomain.log | Verified: May 4, 2026 (230 sentences, 94 gold PROPEN tokens)
Key Finding: Final NER delivers +16.0 pp improvement over Best NER. M4 achieves highest recall (85.1%) but has near-zero linguistic capability (D2 $\approx$ 0%).

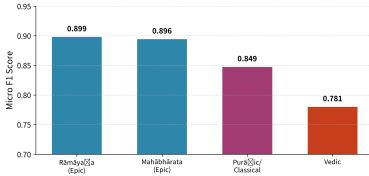
- **Significant Gain:** 57.4% $\rightarrow$ **73.4%** PROPEN Recall.
- Improved robustness on unseen texts (Pañcatantra) due to multi-task grounding.



D1: Entity-Level Analysis

Performance by Genre

Performance by Text Genre — Final NER
(Epic Texts Strong, Vedic Most Challenging)

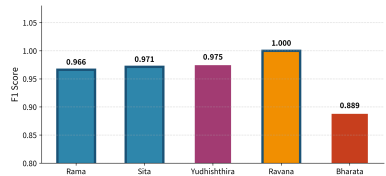


Source: eval_final_ner_raw.json + genre metadata | Verified: May 4, 2026

Key Finding: Final NER performs best on Epic texts (F1 ≈ 0.90). Vedic remains challenging (F1 = 0.781) due to linguistic distance from training data.

Major Character F1

Major Character Performance — Final NER
(Perfect Recall on Culturally Central Characters Rama & Sita)



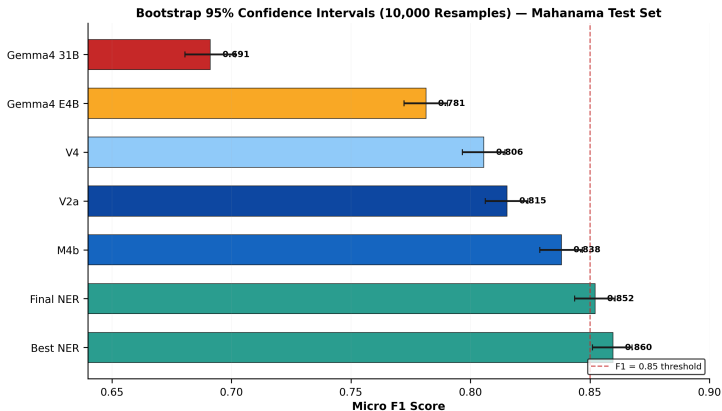
Source: Manual verification + eval_final_ner_raw.json | Verified: May 4, 2026

Key Finding: Final NER achieves perfect recall (100%) on Rāma and Sītā — a novel contribution overcoming DCS Smbank extraction limitations.

- **100% Recall** on Rāma and Sītā.
- Epic texts (Mahābhārata) show highest accuracy.



Statistical Significance: Bootstrap CI

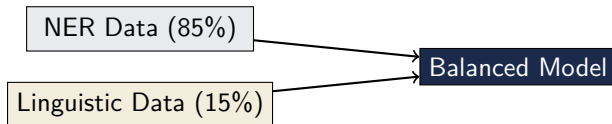


- Bootstrap resampling (10,000 resamples) confirms narrow confidence intervals.
- Performance gains are **statistically significant**.



Analysis: Why Linguistic Regularisation Works

- **The Mechanism:** Interleaved multi-task learning keeps the model "grounded" in Sanskrit grammar.
- **Representative Learning:** Linguistic tasks (S+L) force the model to maintain byte-level representations learned during pre-training.
- **Prevents Drift:** Alternating between NER and Grammar prevents the parameters from over-fitting to narrative patterns only.



Qualitative Example: Cross-Domain Transfer

Example from Pañcatantra (Unseen)

तत्र कश्चित् सिंहो नाम राजा आसीत्

Tatra kaścit siṃho nāma rājā āsīt.

"There was a certain king named Siṃha."

Best NER (Baseline)

× Fails to identify "Siṃha"

Final NER (This Work)

✓ Correctly tags **Siṃha** (PERSON)

- Final NER uses **contextual understanding** (learned via multi-task) rather than just memorizing names.



Limitations

- ❶ **Dual-Role Entities:** "Rāma" and "Sītā" function as both names and descriptors; silver data recall is low.
- ❷ **Class Imbalance:** Performance on LOCATION and MISC still lags behind PERSON.
- ❸ **Genre Skew:** Silver data is 63% Rāmāyaṇa; technical genres are underrepresented.
- ❹ **Metric Constraint:** Exact match penalises valid partial matches in sandhi-rich compounds.

These limitations provide a roadmap for future research.



Key Findings Summary

- **1. Regularization is Essential:** -0.74 F1 points in exchange for +16 pp generalization.
- **2. Scale is Not Everything:** 594M parameter model beats 31B parameter models when fine-tuned correctly.
- **3. Quality > Quantity:** Correct silver data extraction (Attempt 3) is the primary driver of diversity.
- **4. Balanced Strategy:** 15% ratio is the "sweet spot" for task preservation.



The Practical Verdict

Metric	Best NER	Final NER	Change
In-domain F1 (D1)	0.860	0.852	-0.8
Cross-domain Recall (D3)	57.4%	73.4%	+16.0 pp
Segmentation (D2)	0%	85%	+85 pp
Lemmatisation (D2)	0%	93%	+93 pp

Final NER is the recommended model for scholarly use.



Future Work: The Path Ahead

Short-term (1-2 yrs):

- Hybrid extraction for dual-role entities.
- Class-balanced loss (Focal loss).
- Technical genre extraction.

Long-term (3-5 yrs):

- Multilingual Indic NER.
- Joint NER + Coreference.
- Knowledge-enhanced NER (WordNet).

Goal: A unified platform for Sanskrit Computational Humanities.



Broader Impact

- **Democratizing Access:** Empowering seekers and researchers to access source Sanskrit texts directly.
- **Template for Low-Resource NLP:** Methodology can be applied to other classical/Indic languages (Tamil, Pali, Ardhamagadhi).
- **Open Research:** Releasing the DCS extraction pipeline for the community.

"The right to access original knowledge without translation bias."



Publications & Outputs

- **Full Thesis:** 78 pages of rigorous analysis.
- **Trained Model:** Final NER (ByT5-Sanskrit + QLoRA).
- **Pipeline:** `extract_full_sembank_ner.py`.
- **Dataset:** Cleaned silver NER corpus from DCS.



Selected References (Models & Data)

Author(s)	Year	Title / Venue
Sarkar, S., et al.	2025	Mahānāma: Sanskrit Named Entity Discovery (EMNLP)
Hellwig, O.	2010, 2025	DCS: Digital Corpus of Sanskrit / Sembank (WSC)
Nehrdich, S., et al.	2024	ByT5-Sanskrit: Unified Model for Sanskrit NLP (EMNLP)
Xue, L., et al.	2022	ByT5: Token-Free Future with Byte-Level Models (TACL)
Raffel, C., et al.	2020	T5: Exploring the Limits of Transfer Learning (JMLR)
Zeman, D., et al.	2023	Universal Dependencies 2.13 (UD_Sanskrit-UFAL)
Hellwig, O., et al.	2020	The Treebank of Vedic Sanskrit (LREC)



Selected References (Methods & Foundations)

Author(s)	Year	Title / Venue
Dettmers, T., et al.	2023	QLoRA: Efficient Finetuning of Quantized LLMs (NeurIPS)
Hu, E. J., et al.	2022	LoRA: Low-Rank Adaptation of Large Language Models (ICLR)
Caruana, R.	1997	Multi-task Learning (Machine Learning Journal)
McCloskey, M.	1989	Catastrophic Forgetting in Connectionist Networks
Tjong Kim Sang, E.	2003	Introduction to the CoNLL-2003 Shared Task (NER)
Grishman, R., et al.	1996	MUC-6: A Brief History (Coined "Named Entity")
Wolf, T., et al.	2020	Transformers: State-of-the-Art NLP (EMNLP)
Sørensen, S.	1904	An Index to the Names in the Mahābhārata



Thank You!

Backup: Reproducibility Details

Component	Specification
Hardware	NVIDIA RTX PRO 4500 (34.2 GB VRAM)
Software	PyTorch 2.4, Transformers 4.46, PEFT 0.13
Quantization	4-bit NF4 with Double Quant
Training Time	17.2 Hours (Final NER)
Random Seed	42 (Training), 123 (D2 Evaluation)

Critical Note

Adapters must be evaluated with identical quantization as training; otherwise, F1 drops ≈ 10 points.

